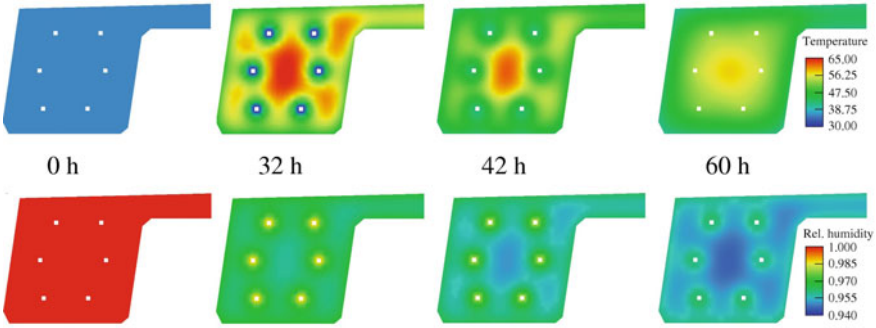


Fig. 7.7 Temperature (top) and relative humidity (bottom) fields in the left symmetric part of the arch cross section. The outer dimensions of the displayed cross section are 3.5 × 2.17 m. Six cooling pipes are embedded, and the cooling water was turned off at 40 h

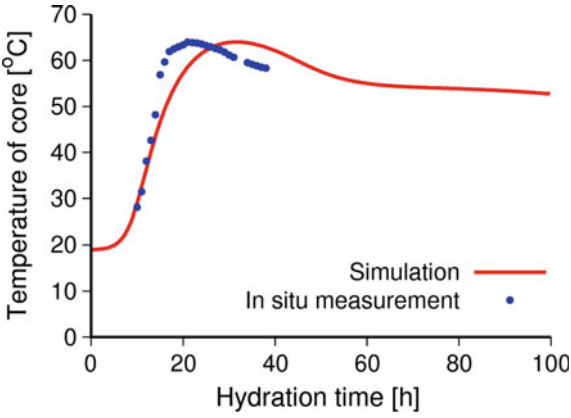


Fig. 7.8 Temperature validation in the core of the bridge arch

maximum temperature would exceed 85 °C, which was unacceptable. Figure 7.8 validates temperatures from the thermo-hygral model.

The results of the transport analysis entered mechanical analysis which computed cracks (see Fig. 7.8 for the evolution of temperature). The cracks' pattern originates from heat dissipation through the surface (Fig. 7.9).

7.3.4.3 Validation of ConCrack's Restrained Beam

The validation of the coupled hydro-thermo-hydro-mechanical analysis originates from the well-documented RG8 ConCrack experiment (Mazars 2014). Figure 7.10 shows the overall geometry of the RG8 beam with two massive heads on both sides and two restraining steel struts. The central beam part was made of reinforced concrete C50/60, $w/c = 0.46$, with dimensions of $0.5 \times 0.8 \times 5.1$ m. The

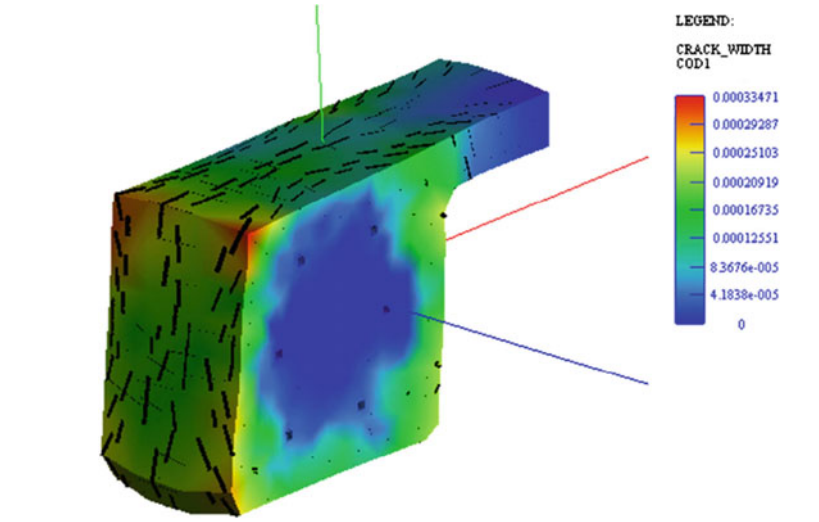


Fig. 7.9 Crack pattern in a 1-m long segment in the Oparno Bridge

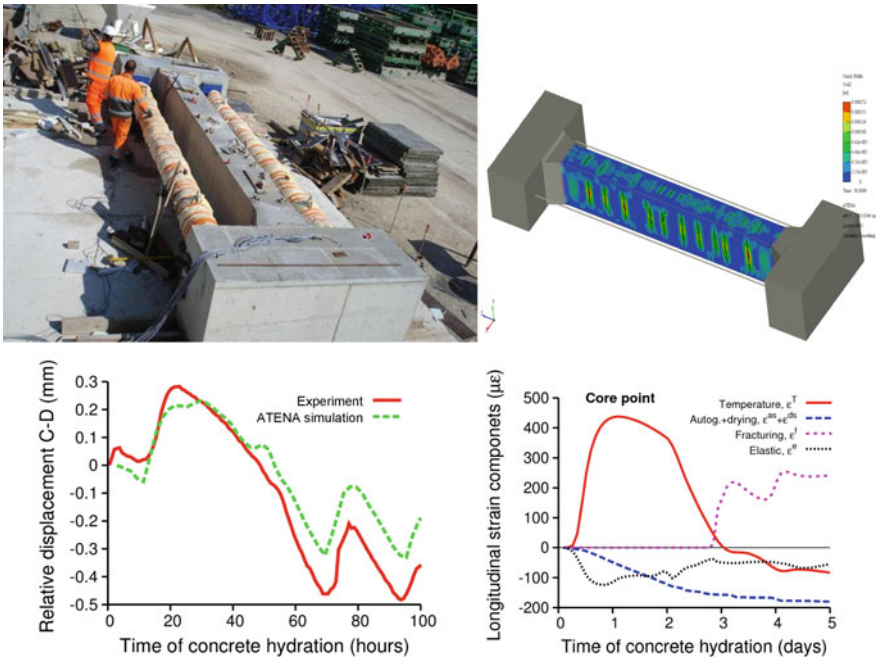


Fig. 7.10 Validation of RG8 ConCrack experiment—experiment overview (by CEOS.fr), cracks at 30 days, displacement of points C-D and strain decomposition at the core point

benchmark experiment provided material and structural data, internal temperature in three points, and displacements of two points C, D located on the beam axis 2.5 m apart. Maximum temperature in the core attained 53.7 °C at 30 h.

Thermo-hygro-mechanical simulation used Künzel's heat-moisture model, B3 creep model and fracture-plastic material model in ATENA software (Červenka et al. 2014). Figure 7.10 validates further C-D displacement and displays strain decomposition in the core point. Note that thermal strain with subsequent cooling contributes the most to cracking at early ages.

7.3.4.4 Validation of a Restrained Wall in Oslo

A restrained wall served as a forerunner experiment for Bjørvika submerged tunnel project, built in Oslo between 2005 and 2012. The wall utilised low-heat concrete with a minimal risk of early-age cracking and high watertightness. Figure 7.11 shows two walls built on a raft concrete slab (Ji 2008). The left wall is made from SV-40 concrete containing low-heat OPC I with silica fume at the amount of 424 kg/m^3. This left wall is validated using a thermo-chemo-mechanical model

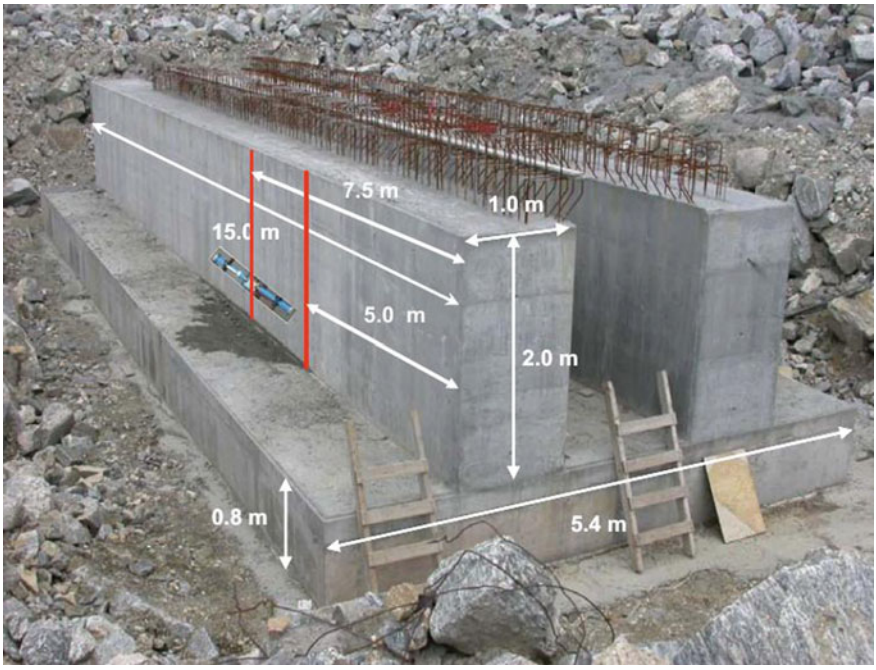


Fig. 7.11 Field test of a restrained wall in Oslo (Ji 2008)